

Newtonian Gravitational Curvature-Induced Entanglement Generation

Mughees Ahmad Khan^{1,*}, Asad Ali^{1,†}, M. I. Hussain¹, H. Kuniyil¹, and Saif Al-Kuwari^{1,‡}

¹*Qatar Center for Quantum Computing, College of Science and Engineering, Hamad Bin Khalifa University, Doha, Qatar*

(Dated: September 1, 2026)

We show that gravitational curvature can control the generation of nonlocal quantum correlations in a hybrid qubit-mechanical device. The tidal field of a nearby source mass modifies the susceptibility of a shared mechanical oscillator, thereby tuning an oscillator-mediated qubit-qubit interaction and the resulting entangling phase. An exact treatment of the dynamics reveals stroboscopic geometric gates whose accumulated phase is directly sensitive to gravitational curvature. Treating the curvature as an unknown parameter, we derive the ultimate quantum limit for its estimation and identify a parity-based measurement that saturates this bound. Solving the full master equation with the mechanical mode retained explicitly, we find that thermal occupation of the mediator suppresses entanglement between the closure times but is undone at each closure, exactly in the unitary limit for any initial mechanical temperature, so that ground-state cooling of the oscillator is not a prerequisite for the protocol. Mechanical damping and qubit dephasing behave differently: they leak branch information irreversibly to the environment, and it is the heating and dephasing rates, rather than the bath occupation alone, that limit the entanglement visibility and the number of usable interrogation loops. In contrast to gravity-mediated entanglement proposals, entanglement is generated by the mechanical oscillator while gravity acts solely as a classical control field. We quantify the achievable curvature parameters; all curvature dependencies are analytic, allowing for exact rescaling of the results. The scheme therefore demonstrates gravitational control of a quantum interaction and provides a route to curvature sensing based on a nonlocal two-qubit phase rather than on local phase measurements.

I. INTRODUCTION

Mechanical oscillators have emerged as sensitive transducers connecting gravitational fields to quantum observables [1, 2]. Their center-of-mass motion couples directly to Newtonian forces, and advances in levitation, cooling, and quantum control have pushed nanoscale resonators into regimes where quantum effects are measurable even under gravitational perturbations [3]. The gravitational field of millimeter-sized source masses has been resolved [4], and a mechanical resonator has been prepared in its quantum ground state and controlled at the single-phonon level [5]. A mechanical resonator is therefore a natural interface at which a gravitational field can potentially be converted into a quantum mechanical observable.

Most quantum gravimetry protocols encode gravitational information in a local observable; an acceleration-induced phase or a frequency shift of a single resonator [6, 7]. The estimator is then built from a single probe, and entanglement plays no role in the encoding; the precision limits of this single-probe approach have been derived at the level of quantum Fisher information [8]. Entanglement and nonlocal correlations can, in principle, improve sensitivity beyond classical limits and extract information inaccessible to single-probe measurements [9–11]. This raises the question of whether a gravitational field can be transduced into a measurable two-body en-

tangling interaction within a hybrid quantum device.

We study a hybrid architecture consisting of two qubits coupled longitudinally to a common mechanical resonator. Longitudinal coupling commutes with the qubit Hamiltonian, admits exact analytical treatment, and produces high-fidelity oscillator-mediated gates [12, 13]. The resonator susceptibility determines the strength of the induced two-qubit Ising interaction; gravitational curvature modifies this susceptibility and thereby controls the rate of entanglement generation. The shared mode is essential because the displacement conditioned on the state of one qubit acts on the other qubit, and eliminating the oscillator leaves a two-qubit interaction. The individual components of geometric phase gates, facilitated by a shared bosonic mode, and the softening of a mechanical frequency due to Newtonian effects are both well-established concepts. This work presents a significant contribution by assembling a fully analytical transduction chain that connects the tidal field to the entangling phase of two qubits. Additionally, it provides a precise analysis of which parts of this chain remain intact despite thermal occupation, mechanical damping, and qubit dephasing.

The mechanism is conceptually distinct from the Bose-Marletto-Vedral (BMV) proposals [14, 15], in which the gravitational field itself is the candidate quantum mediator. In BMV schemes, observation of entanglement may constitute evidence for the quantum nature of gravity, and optomechanical implementations of this idea have been analyzed in detail [16, 17]; gravity-mediated entanglement between two oscillators has been analyzed as a quantum superposition of geometries [18], and the source-mass form factors that optimize gravity-induced

* mukh68937@hbku.edu.qa

† asal68826@hbku.edu.qa

‡ smalkuwari@hbku.edu.qa

entanglement have been derived [19]. In the present work the mechanical oscillator, not gravity, is the explicit quantum mediator: gravity does not transmit quantum information between the qubits but modifies a property of the mediator, its frequency, which enters only as a classical background field. The observation of entanglement here is evidence of gravitational control of a quantum gate, not of graviton exchange; no quantization of the gravitational field is assumed. Related work on gravity-induced decoherence [20], optomechanical gravity signatures [21], and quantum correlations in curved space-time [22] illustrates the breadth of gravitational effects accessible in quantum-mechanical systems.

The gravitational effect of primary interest in our current contribution is not the uniform acceleration due to gravity but rather *the curvature, or tidal field, associated with the spatial gradient of the gravitational force*. The equivalence principle states that a uniform field can be eliminated by transitioning to a freely falling frame, while the tidal field cannot be removed [23, 24]. Uniform acceleration shifts the oscillator equilibrium and produces local single-qubit phases suppressible by a spin-echo sequence [25]; being local unitaries, these phases cannot generate or alter entanglement. Curvature modifies the effective restoring force and mechanical frequency, changing the susceptibility of the shared mode and encoding a nonlocal two-qubit phase; this relative phase between two-qubit computational states is observable. It is the second-order spatial gradient of the Newtonian potential that enters the two-qubit dynamics.

This paper quantifies this mechanism. A source of mass M at a distance R produces a bare curvature $\Gamma_R = GM/R^3$. The linear gravitational force displaces the oscillator equilibrium, so the curvature that acts on the mode is $\Gamma_{\text{eq}} = GM/(R + x_{\text{eq}})^3$, which renormalizes the mechanical frequency from its bare value ω_0 to $\Omega^2 = \omega_0^2 - 2\Gamma_{\text{eq}}$. The renormalized frequency sets the susceptibility of the shared mode and hence the strength of the induced Ising coupling [26, 27]. We treat the qubit-oscillator coupling exactly; a polaron transformation removes the coupling, and the resulting Magnus expansion terminates at second order, so the evolution operator holds at arbitrary coupling strength without rotating-wave or dispersive approximations [28, 29]. At stroboscopic times, the oscillator disentangles from the qubits and leaves a pure two-qubit phase gate, in direct analogy with the geometric phase gates of trapped-ion quantum computing [30, 31]. We then retain the mechanical mode explicitly in a Markovian master equation with mechanical damping, a thermal bath, and qubit dephasing. We show that the effect of an initially thermal mediator is undone exactly at the closure times, whereas damping and dephasing leak information about the qubit state to the environment and limit the usable interrogation time. Finally, treating the curvature as an unknown parameter of the closed-system state at closure, we derive the quantum Fisher information, propose a parity-measurement readout that saturates the quantum Cramér-Rao bound, and

add a differential source protocol; the Fisher-information analysis of the open-system state is left for future work.

The paper is organized as follows. Section II introduces the Hamiltonian and the curvature-dependent mechanical frequency. Section III derives the effective qubit-qubit interaction. Section IV analyzes the entanglement dynamics and the stroboscopic entangling gate. Section V treats the open-system dynamics with the mechanical mode retained explicitly and separates the reversible effect of an initially thermal oscillator from the irreversible effects of mechanical damping and qubit dephasing. Section VI formulates the curvature estimation problem for the closed-system state at closure, derives the quantum Fisher information, and presents the parity-based readout together with [the spin-echo-assisted differential sensing scheme](#). Section VII concludes, and the detailed derivations are provided in Appendices A–F.

II. MODEL AND GRAVITATIONAL RENORMALIZATION

A. Hamiltonian

Our system consists of two qubits with transition frequencies $\omega_{q,i}$ ($i = 1, 2$) coupled longitudinally to a harmonic mechanical mode of bare frequency ω_0 and mass m as shown in Fig. 1. A source mass M is at distance R from the mechanical equilibrium. The total Hamiltonian is

$$H = \hbar\omega_0 b^\dagger b + \sum_{i=1}^2 \frac{\hbar\omega_{q,i}}{2} Z_i + \hbar \sum_{i=1}^2 \lambda_i Z_i (b + b^\dagger) + V_G(x), \quad (1)$$

where $\hbar$ is the reduced Planck constant, b ($b^\dagger$) is the annihilation (creation) operator of the phonon mode, ω_0 and $\omega_{q,i}$ are the angular frequencies of the oscillator and of the qubits, Z_i is the Pauli- z operator of qubit i with $Z_i|0\rangle_i = +|0\rangle_i$, λ_i is the longitudinal coupling rate, $V_G(x) = -GMm/(R + x)$ is the Newtonian interaction with G Newton's constant, and $x = x_{\text{zpf}}(b + b^\dagger)$ is the displacement of the oscillator along the line joining it to the source, with x_{zpf} the zero-point amplitude given below Eq. (3). The longitudinal coupling satisfies $[Z_i, H_q + H_{\text{int}}] = 0$, where H_q and H_{int} denote the second and third terms of Eq. (1), so no excitation is exchanged with the oscillator; instead, each qubit state exerts a different force, of magnitude $\hbar\lambda_i/x_{\text{zpf}}$, on the resonator. The qubit frequencies drop out of every result below because Z_i commutes with the rest of H . Three assumptions are built into Eq. (1) and used throughout; the oscillator moves along a single axis pointing at the source, the source is static during each interrogation, and the qubits couple only through Z_i (no transverse component); the last is what makes the model exactly solvable in Sec. III. Apart from the expansion of V_G to second order in x , made in Sec. II B, no further approximation

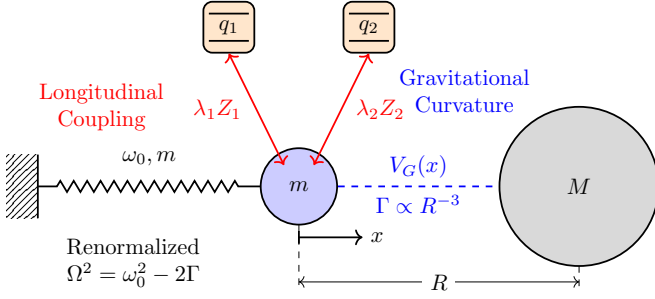


FIG. 1. Schematic of the proposed setup. A mechanical oscillator of mass m and bare frequency ω_0 is placed at a distance R from a source mass M . The Newtonian gravitational potential $V_G(x)$ induces a curvature $\Gamma \propto R^{-3}$ which renormalizes the mechanical frequency to $\Omega^2 = \omega_0^2 - 2\Gamma$. Two qubits (q_1, q_2) are coupled longitudinally to the oscillator with strengths λ_1, λ_2 . The gravitational curvature controls the effective susceptibility of the oscillator, thereby modulating the mediated two-qubit entanglement rate. The displacement x is measured along the line joining the oscillator and the source.

enters the closed-system treatment of Secs. II–IV and VI; dissipation is added in Sec. V.

B. Frequency renormalization

Expanding $V_G(x)$ through Taylor series for $|x| \ll R$ as shown in Appendix A, one can get

$$V_G(x) \approx \text{const} + mg_R x - m\Gamma_R x^2 + \mathcal{O}(x^3), \quad (2)$$

where $g_R = GM/R^2$ is the gravitational acceleration of the source at the oscillator, $\Gamma_R = GM/R^3$ is the bare tidal curvature, and the constant, $-GMm/R$, only fixes the energy reference and is dropped below. The linear term displaces the equilibrium to $x_{\text{eq}} \simeq -g_R/\omega_0^2$, and the quadratic term reduces the effective stiffness of the oscillator: the side nearer to the source is pulled more strongly than the far side. The neglected cubic term is smaller than the quadratic one by a factor of order x/R , below 10^{-6} for subnanometer displacements and millimeter distances. Expanding about the true equilibrium yields the gravity-renormalized frequency

$$\Omega^2 = \omega_0^2 - 2\Gamma_{\text{eq}}, \quad (3)$$

where $\Gamma_{\text{eq}} = GM/(R + x_{\text{eq}})^3$ is the curvature at the displaced equilibrium; below we write Γ for Γ_{eq} wherever the distinction from Γ_R is immaterial. Equation (3) is the stiffness at the displaced equilibrium and is exact for a pointlike or spherical source within the harmonic approximation. The dimensionless curvature parameter $\epsilon \equiv \Gamma_{\text{eq}}/\omega_0^2$ gives $\Omega = \omega_0\sqrt{1 - 2\epsilon}$; the mode is stable for $\epsilon < 1/2$. In Fig. 2, ϵ is set to 0–0.05 for illustration; these values are chosen for visibility. The zero-point displacement, $x_{\text{zpf}} = (\hbar/2m\Omega)^{1/2}$, is the amplitude of the

ground-state phonon mode and grows as the mode softens. Note that due to the traceless nature of the Newtonian tidal tensor in vacuum, transverse mechanical modes are correspondingly stiffened by $+\Gamma_{\text{eq}}$ (Appendix B), a unique tensor signature of the gravitational field.

C. Force and curvature channels

The uniform acceleration g_R displaces the equilibrium and produces local qubit-frequency shifts proportional to x_{eq} , and hence to g_R . They arise because writing $x = x_{\text{eq}} + X$ in the coupling term of Eq. (1) shifts $b + b^\dagger$ by the constant $x_{\text{eq}}/x_{\text{zpf}}$; being local unitaries, they are absorbed into $\omega_{q,i}$ and canceled by the echo sequence of Sec. VI. The curvature Γ_{eq} modifies Ω and the mechanical susceptibility $\chi_m = 1/m\Omega^2$, the static displacement per unit force. Since the oscillator-mediated qubit-qubit interaction depends on χ_m , curvature is the channel encoding the nonlocal two-qubit phase.

After shifting to the fluctuation coordinate $X = x - x_{\text{eq}}$, retaining the harmonic approximation, the model becomes

$$\frac{H}{\hbar} = \Omega b^\dagger b + \sum_{i=1}^2 \frac{\omega_{q,i}}{2} Z_i + \sum_{i=1}^2 \lambda_i Z_i (b + b^\dagger), \quad (4)$$

where all parameters are evaluated at the gravitationally shifted equilibrium, and b now denotes the annihilation operator of the renormalized mode of frequency Ω , so that x_{zpf} , and in the microscopic model of Sec. III, the couplings λ_i , depend on Ω . The first two terms constitute the free Hamiltonian H_0 , and the last term is the qubit-oscillator coupling. Eq. (4) is thus Eq. (1) with the linear force absorbed into x_{eq} and the local qubit shifts, the curvature absorbed into Ω , and the constant $V_G(x_{\text{eq}})$ dropped.

III. EFFECTIVE QUBIT-QUBIT INTERACTION

A. Polaron transformation

The Hamiltonian (4) is exactly diagonalizable via a spin-dependent displacement. We define the collective force operator $S = \lambda_1 Z_1 + \lambda_2 Z_2$ and the polaron unitary $U_P = \exp[(b^\dagger - b)S/\Omega]$. Because $[S, b] = [S, Z_i] = 0$, the Baker-Campbell-Hausdorff series terminates at first order, giving $U_P b U_P^\dagger = b - S/\Omega$. The linear coupling then cancels exactly (Appendix C), yielding

$$H' = \hbar\Omega b^\dagger b + \sum_{i=1}^2 \frac{\hbar\omega_{q,i}}{2} Z_i - \hbar J Z_1 Z_2, \quad (5)$$

in which the oscillator is completely decoupled and

$$J(\Gamma_{\text{eq}}) = \frac{2\lambda_1(\Gamma_{\text{eq}})\lambda_2(\Gamma_{\text{eq}})}{\Omega(\Gamma_{\text{eq}})}. \quad (6)$$

The transformation leaves a residual term $-\hbar S^2/\Omega$ (Appendix C), the energy gained as the oscillator relaxes to its force-dependent equilibrium. Expanding S^2 splits this into two parts. The third term in Eq. (5) is the Ising coupling $-\hbar J Z_1 Z_2$, with $J = 2\lambda_1 \lambda_2/\Omega$ the oscillator-mediated coupling strength between the qubits. The identity part, $-\hbar(\lambda_1^2 + \lambda_2^2)/\Omega$, which is a global energy shift is dropped. No rotating-wave, dispersive, or weak-coupling approximation is required. The coupling λ_i/Ω may be arbitrary, and the result rests only on the assumptions of Sec. II and on the absence of dissipation, lifted in Sec. V.

B. Curvature dependence

Two models for the coupling are relevant. They differ only in how λ_i responds when curvature shifts the mode frequency Ω through Eq. (3), that is, through the dimensionless curvature parameter $\epsilon = \Gamma_{\text{eq}}/\omega_0^2$.

In the first model the couplings are stabilized. The λ_i are held at their zero-curvature values $\lambda_i^{(0)}$ by calibration or feedback while Ω varies. Only the explicit Ω^{-1} of Eq. (6) then carries the curvature dependence. The stabilized coupling therefore obeys $J_{\text{st}} \propto \Omega^{-1}$, so that $J_{\text{st}}/J_{\text{st}}(0) = (1 - 2\epsilon)^{-1/2}$, where $J_{\text{st}}(0)$ is its value at $\epsilon = 0$. To first order in ϵ the fractional sensitivity is $\delta J_{\text{st}}/J_{\text{st}}(0) \simeq \epsilon$, with $\delta J_{\text{st}} = J_{\text{st}} - J_{\text{st}}(0)$ the curvature-induced change. This is the model evaluated in Fig. 2, where $\lambda_1 = \lambda_2$ are held fixed while Ω varies with ϵ . Any uncompensated drift of $\lambda_1 \lambda_2$ is then indistinguishable from a curvature signal of the same fractional size.

In the second model the coupling is microscopic. Each qubit couples to the oscillator through the gradient of its transition energy, so that $\lambda_i = \varkappa_i x_{\text{zpf}}/2\hbar$. Here $\varepsilon_i(x)$ is the transition energy of qubit i as a function of the oscillator position, and $\varkappa_i = \partial \varepsilon_i / \partial x|_{x_{\text{eq}}}$ is its gradient at equilibrium. The quantity held fixed is now $\varkappa_i$, not λ_i . Since $x_{\text{zpf}} = (\hbar/2m\Omega)^{1/2}$, we have $\lambda_i \propto \Omega^{-1/2}$ and hence $\lambda_1 \lambda_2 \propto \Omega^{-1}$. The microscopic coupling therefore scales as $J_{\text{mic}} \propto \Omega^{-2}$, which gives $J_{\text{mic}}/J_{\text{mic}}(0) = (1 - 2\epsilon)^{-1}$ and a fractional sensitivity $\simeq 2\epsilon$, twice the stabilized-coupling result. The enhancement arises because curvature modifies both the mechanical susceptibility and the zero-point amplitude, each entering the coupling independently. Equivalently, inserting $x_{\text{zpf}}^2 = \hbar/2m\Omega$ into Eq. (6) gives $J = \varkappa_1 \varkappa_2 \chi_m / 4\hbar$ with $\chi_m = 1/m\Omega^2$, so that J is the product of the two force gradients and the static susceptibility.

The general logarithmic sensitivity $J^{-1} dJ/d\Gamma_{\text{eq}}$ follows from Eqs. (3) and (6). Differentiating $\Omega^2 = \omega_0^2 - 2\Gamma_{\text{eq}}$ gives $d\Omega/d\Gamma_{\text{eq}} = -1/\Omega$. For $J_{\text{st}} \propto \Omega^{-1}$ this yields $J^{-1} dJ/d\Gamma_{\text{eq}} = 1/\Omega^2$, and for $J_{\text{mic}} \propto \Omega^{-2}$ it yields $2/\Omega^2$. Both reduce to the fractional sensitivities ϵ and 2ϵ quoted above, because $\Omega^2 \rightarrow \omega_0^2$ and $\Gamma_{\text{eq}}/\omega_0^2 \rightarrow \epsilon$ in the small-curvature limit.

The two models are the limits $\eta_i = 0$ and $\eta_i = -1/2$ of a single expression, where $\eta_i \equiv d \ln \lambda_i / d \ln \Omega$ measures

how the coupling responds to a change in the mode frequency. In general $J^{-1} dJ/d\Gamma_{\text{eq}} = (1 - \eta_1 - \eta_2)/\Omega^2$.

The curvature-transduction chain is

$$\Gamma_{\text{eq}} \rightarrow \Omega(\Gamma_{\text{eq}}) \rightarrow \lambda_i(\Gamma_{\text{eq}}) \rightarrow J(\Gamma_{\text{eq}}) \rightarrow \phi_{12}(\Gamma_{\text{eq}}), \quad (7)$$

where ϕ_{12} is the phase accumulated on $Z_1 Z_2$. It equals θ_k at the closure times of Sec. IV. Each arrow is an identity of the harmonic model and carries no further approximation.

IV. ENTANGLEMENT DYNAMICS

A. Exact laboratory-frame propagator

In the interaction picture with respect to the free Hamiltonian, we conjugate the Schrödinger-picture coupling $\hbar S(b + b^\dagger)$ with the free propagator $U_0(t) = e^{-iH_0 t/\hbar}$, giving $H_I(t) = U_0^\dagger(t) \hbar S(b + b^\dagger) U_0(t)$. The operator S commutes with H_0 and is therefore left unchanged, while the mode operators rotate as $U_0^\dagger(t) b U_0(t) = b e^{-i\Omega t}$ and its Hermitian conjugate, so the coupling reads $H_I(t) = \hbar S(b e^{-i\Omega t} + b^\dagger e^{i\Omega t})$. The propagator then terminates exactly at second Magnus order (Appendix D), because the commutator $[H_I(t_1), H_I(t_2)] \propto S^2$ commutes with all subsequent commutants:

$$U_I(t) = D[\alpha(t)S] \exp[i\Phi(t)S^2], \quad (8)$$

where $D[\beta] = \exp(\beta b^\dagger - \beta^* b)$ is the displacement operator, $\alpha(t) = (1 - e^{i\Omega t})/\Omega$ is the conditional displacement amplitude, and $\Phi(t) = (\Omega t - \sin \Omega t)/\Omega^2$ is the geometric phase function. For a computational state $|s_1 s_2\rangle$, with $s_i = \pm 1$ the eigenvalues of Z_i and $S_s = s_1 \lambda_1 + s_2 \lambda_2$ the corresponding eigenvalue of S , the mechanical mode traces a circle of radius $|S_s|/\Omega$ in phase space. Eq. (8) is thus a qubit-state-dependent displacement times a qubit-state-dependent phase, $\Phi(t)S_s^2$ being proportional to the phase-space area swept by the branch $|s_1 s_2\rangle$; this is the structure of the Mølmer-Sørensen and geometric phase gates of trapped ions [30, 31], realized here in the limit in which the state-dependent force is static in the laboratory frame; closely related geometric phase gates based on ac-Stark shifts have been formulated for cavity-QED platforms [32, 33]. In the trapped-ion gates, the phase-space loop is driven by a bichromatic field and closes at the inverse detuning; here the coupling in Eq. (4) carries no explicit time dependence, the residual rotation in the interaction picture occurs at $t_k = 2\pi k/\Omega$.

B. Stroboscopic closure

The trajectories close, $\alpha(t_k) = 0$, at $t_k = 2\pi k/\Omega$ for $k = 1, 2, 3, \dots$. At these times, the displacement operator

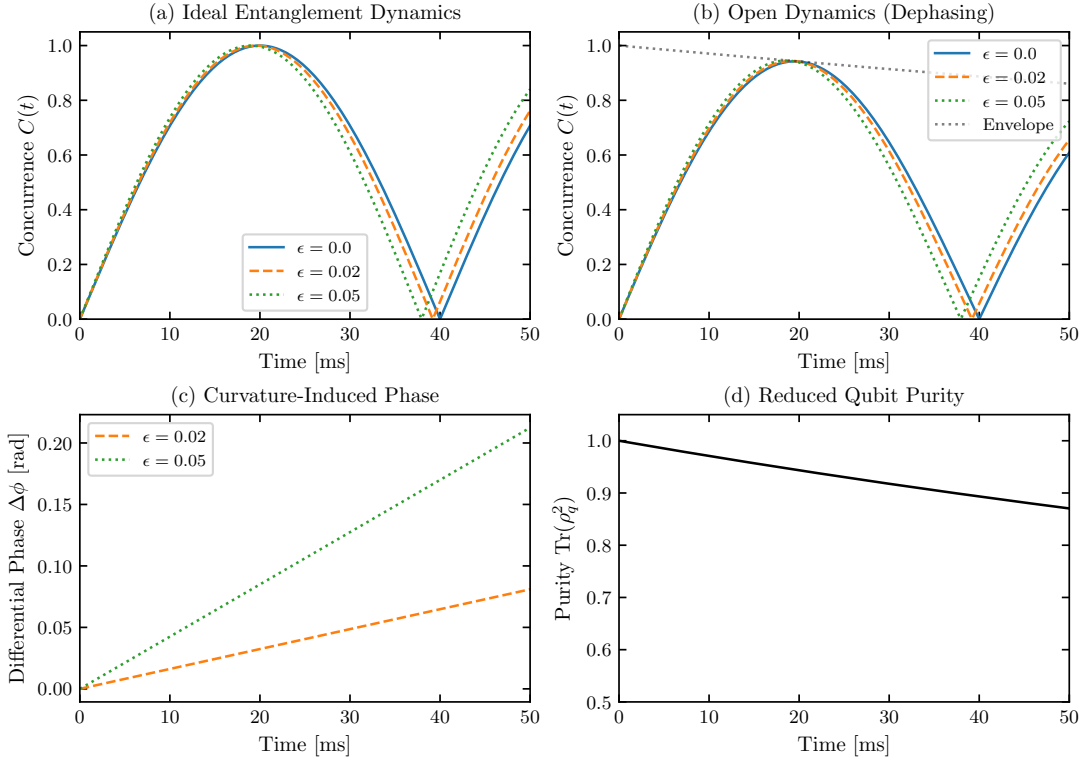


FIG. 2. Effective-model entanglement dynamics as a function of gravitational curvature. (a) Two-qubit concurrence for $\epsilon = 0, 0.02$, and 0.05 with no dephasing. Increasing ϵ shifts the oscillation via the enhanced susceptibility. (b) Same as (a) with independent qubit dephasing $\gamma_\phi = 1.5 \text{ s}^{-1}$; the envelope decays as $e^{-2\gamma_\phi t}$. (c) Differential entangling phase $\Delta\phi = 2t[J(\epsilon) - J(0)]$ versus time, showing linear accumulation of the curvature-induced phase difference. (d) Reduced two-qubit purity $\text{Tr}(\rho_q^2)$ under qubit dephasing only. Purity curves for the three ϵ values overlap in this panel because the dephasing channel is ϵ -independent in the effective model; the small spread becomes visible at late times due to the ϵ -dependent oscillation frequency. Parameters arbitrarily set are $\omega_0/2\pi = 200 \text{ Hz}$, $\lambda/2\pi = 25 \text{ Hz}$, $J(0)/2\pi \simeq 6.25 \text{ Hz}$.

becomes the identity, and the oscillator disentangles from the qubits exactly, regardless of the initial mechanical state (Sec. VB makes this explicit for a thermal initial state). The Schrödinger-picture mechanical state returns to its initial condition since $e^{-i\Omega t_k b^\dagger b} = \mathbf{1}$ for integer k . As a result, the mechanical mode factors out of the joint state. The qubits remain in a pure two-qubit state, disentangled from the oscillator, while the qubit-qubit correlations generated during the loop are retained. The accumulated two-qubit gate is

$$U_{\text{ent}}(t_k) = \exp[i\theta_k Z_1 Z_2], \quad \theta_k = \frac{4\pi k \lambda_1 \lambda_2}{\Omega^2} = J t_k, \quad (9)$$

where θ_k is the qubit-state-dependent geometric phase after k complete loops projected onto $Z_1 Z_2$, the global phase $\Phi(t_k)(\lambda_1^2 + \lambda_2^2)$ has been dropped (Appendix D), and the second equality confirms consistency with the polaron-frame Hamiltonian. Closure does require the loop time to be set at the curvature-shifted frequency Ω , a mistimed loop leaves a residual displacement $\alpha(t)S$ and hence residual qubit-oscillator correlation.

C. Concurrence at closure

For the initial separable state $|+\rangle_1|+\rangle_2$, with $|+\rangle_i = (|0\rangle_i + |1\rangle_i)/\sqrt{2}$, the Wootters concurrence [34] at a closure time is

$$C(t_k) = |\sin(2\theta_k)| = |\sin(2J t_k)|. \quad (10)$$

Maximum entanglement ($C = 1$) occurs when $\theta_k = \pi/4 + n\pi/2$, with n a non-negative integer. Both J and t_k carry the curvature dependence of $\Omega(\Gamma_{\text{eq}})$. The closure phase θ_k therefore encodes Γ_{eq} . It does so with higher sensitivity than a fixed-time protocol, because the closure time lengthens as the mode softens. Here $t_k \propto \Omega^{-1}$, so Eq. (9) gives $\theta_k \propto \Omega^{-2}$ for stabilized couplings and $\theta_k \propto \Omega^{-3}$ for microscopic couplings. The fractional sensitivity $\theta_k^{-1} \partial_{\Gamma_{\text{eq}}} \theta_k$ is then $2/\Omega^2$ and $3/\Omega^2$, respectively. For a phase Jt accumulated over a fixed time t , it is $1/\Omega^2$ and $2/\Omega^2$.

Fig. 2 is computed from the effective Hamiltonian of Eq. (5), after the oscillator has been eliminated. It is evaluated at a fixed time, not at the closure times. Panel (a) plots $C(t) = |\sin 2J(\epsilon)t|$ for $\epsilon = 0, 0.02$, and 0.05 . Curvature enters only through $J(\epsilon)$. Its fractional

change $\simeq \epsilon$ shifts the first maximum, at $t = \pi/4J(0)$, by $\simeq -\epsilon t$. Panel (c) plots the same effect as the phase difference $\Delta\phi = 2t[J(\epsilon) - J(0)]$. This phase is linear in t because J is time-independent. Panels (b) and (d) include qubit dephasing phenomenologically. The concurrence is multiplied by the envelope $e^{-2\gamma_\phi t}$, which is a two-qubit coherence that decays at twice the single-qubit dephasing rate γ_ϕ . The reduced purity $\text{Tr}\rho_q^2$ of the two-qubit state ρ_q follows from the same dephasing model: the four computational-basis populations are constant, the eight coherences with a single flipped qubit decay at γ_ϕ , and the four with both qubits flipped decay at $2\gamma_\phi$, giving $\text{Tr}\rho_q^2 = \frac{1}{4}(1 + e^{-2\gamma_\phi t})^2$, which decays from unity toward the fully dephased value $1/4$. Both are ϵ -independent by construction, so panel (d) contains a single curve. The stroboscopic result of Eq. (9) coincides with panel (a) at the closure times $t_k = 2\pi k/\Omega$. For the parameters of the figure, $t_1 = 1/(200\text{ Hz}) = 5\text{ ms}$. The first maximum, $C = 1$ at $t = \pi/4J(0) = 20\text{ ms}$, is the closure $k = 4$, with $\theta_4 = \pi/4$. The $\epsilon = 0$ curve is the control for the curvature effect. The signal is the phase difference $\Delta\phi$, not the presence of entanglement, which is generated equally at $\epsilon = 0$.

In summary, gravitational curvature is transduced into an observable two-qubit entangling phase through the oscillator-mediated interaction. At the stroboscopic closure times t_k the mechanical mode disentangles exactly from the qubits. This leaves the pure state $|\psi(\theta_k)\rangle = e^{i\theta_k Z_1 Z_2}|++\rangle$, with concurrence $C(t_k) = |\sin(2\theta_k)|$. Both J and t_k inherit the curvature dependence of $\Omega(\Gamma)$, so the accumulated phase $\theta_k = Jt_k$ carries information about the tidal field. Concurrence quantifies the amount of entanglement generated. It does not characterize how precisely the curvature Γ can be inferred from the state. This motivates a quantum-metrological analysis, in which Γ is treated as an unknown parameter encoded in $\theta_k(\Gamma)$. In Sec. VI, we quantify the ultimate sensitivity of the closed-system protocol using quantum Fisher information, after demonstrating in Sec. V that the stroboscopic mechanism remains effective despite realistic noise. Furthermore, we identify a measurement that achieves the corresponding quantum limit.

V. OPEN-SYSTEM DYNAMICS

The preceding sections establish the exact unitary evolution of the qubit-oscillator system. A realistic mechanical mode, however, exchanges energy with a thermal reservoir, while the qubits experience phase noise. These processes must be included to determine whether the stroboscopic disentangling mechanism of Sec. IV survives under experimentally relevant conditions.

Throughout this section we retain the mechanical mode explicitly rather than replacing it by the effective Ising interaction of Eq. (5). This is essential away from the closure times, where the qubits and the oscillator are generally correlated, and it permits a direct distinction

between three physically different effects: reversible dephasing caused by an initially mixed mechanical state, irreversible information leakage caused by mechanical damping, and direct qubit dephasing.

A. Markovian master equation

Under the Born–Markov and secular approximations, the density operator of the two qubits and the mechanical mode satisfies

$$\begin{aligned} \dot{\rho} = & -\frac{i}{\hbar}[H, \rho] + \kappa_m(\bar{n}_{\text{th}} + 1)\mathcal{D}[b]\rho + \kappa_m\bar{n}_{\text{th}}\mathcal{D}[b^\dagger]\rho \\ & + \sum_{i=1}^2 \frac{\gamma_{\phi,i}}{2}\mathcal{D}[Z_i]\rho, \end{aligned} \quad (11)$$

where H is the Hamiltonian of Eq. (4) and $\mathcal{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}L^\dagger L\rho - \frac{1}{2}\rho L^\dagger L$ is the Lindblad dissipator for a jump operator L . The mechanical dissipators are those in the undriven mode; the static force in Eq. (4) displaces the oscillator without altering its coupling to a bath with a flat spectral density around Ω , as required by the Markov approximation. Here κ_m is the mechanical energy-decay rate and

$$\bar{n}_{\text{th}} = [\exp(\hbar\Omega/k_B T_m) - 1]^{-1} \quad (12)$$

the equilibrium occupation of a mechanical bath at temperature T_m , with k_B the Boltzmann constant, evaluated at the curvature-renormalized frequency of Eq. (3). The rate $\gamma_{\phi,i}$ describes pure dephasing of qubit i ; it is the microscopic origin of the phenomenological envelope used in Fig. 2. The first mechanical dissipator describes phonon loss into the bath and the second thermally stimulated absorption; together they drive the oscillator toward the thermal state with mean occupation $\bar{n}_{\text{th}}$. The qubit dissipator leaves computational-basis populations unchanged and attenuates only coherences between states with different Z_i eigenvalues (Appendix E). We first isolate the effect of an initially thermal oscillator by setting $\kappa_m = \gamma_{\phi,i} = 0$; mechanical damping and qubit dephasing are then treated separately.

B. Reduced qubit dynamics for an initially thermal oscillator

Let the initial state be $\rho(0) = \rho_q(0) \otimes \rho_m^{\text{th}}(\bar{n}_0)$ with $\rho_q(0) = |++\rangle\langle ++|$ and $\rho_m^{\text{th}}(\bar{n}_0)$ the thermal state of the mechanical mode with mean occupation $\bar{n}_0$, written out in Appendix E. We stress that $\bar{n}_0$ is the occupation of the *initial* mechanical state and is in general distinct from the bath occupation $\bar{n}_{\text{th}}$ of Eq. (12).

Applying the exact propagator of Eq. (8) to this state and tracing out the oscillator (Appendix E) shows that each coherence $\rho_{\mathbf{s}\mathbf{s}'}^{(q)} = \langle \mathbf{s} | \text{Tr}_m \rho | \mathbf{s}' \rangle$ of the reduced qubit state, where Tr_m is the trace over the mechanical mode,

between the eigenstates $|\mathbf{s}\rangle = |s_1, s_2\rangle$ of Z_1 and Z_2 , with $S_{\mathbf{s}} = s_1\lambda_1 + s_2\lambda_2$ the corresponding eigenvalue of S , acquires the phase $e^{i\Phi(t)(S_{\mathbf{s}}^2 - S_{\mathbf{s}'}^2)}$ and a real factor $\chi_{\mathbf{s}\mathbf{s}'}(t) = \text{Tr}_m[D(\xi)\rho_m^{\text{th}}]$ with $\xi = \alpha(t)\Delta S_{\mathbf{s}\mathbf{s}'}$ and $\Delta S_{\mathbf{s}\mathbf{s}'} \equiv S_{\mathbf{s}} - S_{\mathbf{s}'}$. This trace is the characteristic function of the thermal state, and the thermal attenuation of the qubit coherence, $\chi_{\mathbf{s}\mathbf{s}'}(t) \equiv \mathcal{C}_{\mathbf{s}\mathbf{s}'}^{\text{th}}(t)$, is

$$\mathcal{C}_{\mathbf{s}\mathbf{s}'}^{\text{th}}(t) = \exp\left[-\frac{2\bar{n}_0 + 1}{2} |\alpha(t)|^2 (\Delta S_{\mathbf{s}\mathbf{s}'})^2\right]. \quad (13)$$

The conditional displacement amplitude of Eq. (8) satisfies

$$|\alpha(t)|^2 = \frac{(1 - e^{i\Omega t})(1 - e^{-i\Omega t})}{\Omega^2} = \frac{4\sin^2(\Omega t/2)}{\Omega^2}, \quad (14)$$

so that at the closure times $t_k = 2\pi k/\Omega$ of Eq. (9) one has $e^{i\Omega t_k} = 1$, $\alpha(t_k) = 0$, and therefore

$$\mathcal{C}_{\mathbf{s}\mathbf{s}'}^{\text{th}}(t_k) = 1 \quad \text{for every } \bar{n}_0. \quad (15)$$

Equation (15) is exact for unitary evolution; an initially thermal mechanical state suppresses qubit coherence between closure times but causes no residual loss at an exact closed-loop time. Between closures the suppression is strongest for the coherences with the largest $|\Delta S_{\mathbf{s}\mathbf{s}'}|$. For $\lambda_1 = \lambda_2 = \lambda$, the $|00\rangle$ – $|11\rangle$ coherence, with $|\Delta S_{\mathbf{s}\mathbf{s}'}| = 4\lambda$, is attenuated most, whereas the $|01\rangle$ – $|10\rangle$ coherence has $\Delta S_{\mathbf{s}\mathbf{s}'} = 0$, because both branches exert the same force on the oscillator, and is never affected by it. The entangling gate of Eq. (9), the concurrence Eq. (10), and consequently the curvature estimation of Sec. VI, which uses the state at closure, are therefore independent of the initial mechanical temperature. This is the curvature-sensing counterpart of the thermal robustness of the Mølmer–Sørensen gate [30], and it removes ground-state cooling of the mediator from the list of prerequisites for the protocol. It should not, however, be confused with immunity to mechanical damping, which transfers information irreversibly to the external bath.

C. Mechanical damping and environmental information leakage

For a fixed qubit configuration $\mathbf{s}$ the oscillator is driven by the c-number force $S_{\mathbf{s}}$. Its conditional first moment $\beta_{\mathbf{s}}(t) = \text{Tr}_m[b\rho_{\mathbf{s}}(t)]$, where $\rho_{\mathbf{s}}(t) = \langle \mathbf{s} | \rho(t) | \mathbf{s} \rangle / p_{\mathbf{s}}$ is the mechanical state conditioned on $\mathbf{s}$ and $p_{\mathbf{s}}$ its population, which is constant because Z_i commutes with H and with every dissipator, follows from Eq. (11) (Appendix E). For two branches that start from a common mechanical state, the separation $\Delta\beta = \beta_{\mathbf{s}} - \beta_{\mathbf{s}'}$ has the squared magnitude

$$|\Delta\beta(t)|^2 = \frac{(\Delta S_{\mathbf{s}\mathbf{s}'})^2}{\Omega^2 + \kappa_m^2/4} \left[1 + e^{-\kappa_m t} - 2e^{-\kappa_m t/2} \cos \Omega t\right]. \quad (16)$$

Damping thus deforms the trajectory itself. In particular, at the undamped closure times,

$$|\Delta\beta(t_k)|^2 = \frac{(\Delta S_{\mathbf{s}\mathbf{s}'})^2}{\Omega^2 + \kappa_m^2/4} \left(1 - e^{-\kappa_m t_k/2}\right)^2, \quad (17)$$

which is nonzero for finite κ_m : the exact geometric closure established in Sec. IV is a property of the unitary dynamics, and with damping the nominal closure is only approximate. The residual displacement is small: $(1 - e^{-\kappa_m t_k/2})^2 \simeq (\kappa_m t_k/2)^2$ is of second order in $\kappa_m t_k$. It enters the qubit coherence through Eq. (13), with $|\alpha(t)|^2 (\Delta S_{\mathbf{s}\mathbf{s}'})^2$ replaced by $|\Delta\beta(t)|^2$. The first-order effect of damping at closure is the loss of information to the bath, derived next.

We next derive the decoherence caused by information emitted into the mechanical reservoir. Over an infinitesimal interval dt , mechanical damping acts as a thermal attenuation channel that mixes the mechanical mode with a thermal bath mode (Appendix E). The two conditional amplitudes $\beta_{\mathbf{s}}$ and $\beta_{\mathbf{s}'}$ leave the bath mode in two states whose overlap is its thermal characteristic function, evaluated at the separation $\sqrt{\kappa_m dt} \Delta\beta(t)$. Independent bath increments multiply, so the visibility accumulated over $[0, t]$ is $e^{-\Lambda(t)}$ with

$$\Lambda(t) = \frac{\kappa_m}{2} (2\bar{n}_{\text{th}} + 1) \int_0^t |\Delta\beta(\tau)|^2 d\tau. \quad (18)$$

The environment therefore distinguishes the two qubit branches in proportion to the squared separation of their conditional mechanical trajectories. Eq. (18) is not only a leading-order estimate: for the Markovian model of Eq. (11) it is exact, because in the Heisenberg picture every bath mode is displaced by an amount proportional to $S_{\mathbf{s}}$, and the squared distance between the bath states of two branches, summed over all bath modes, is $\kappa_m \int_0^t |\Delta\beta(\tau)|^2 d\tau$. Combining the two mechanisms, a qubit coherence at time t has magnitude $\exp[-\frac{1}{2}(2\bar{n}_0 + 1)|\Delta\beta(t)|^2 - \Lambda(t)]$ when $\gamma_{\phi,i} = 0$; the first factor is Eq. (13) evaluated on the damped trajectory, and the second is the loss to the bath.

In the weak-damping regime $\kappa_m/\Omega \ll 1$, and for times $\kappa_m t \ll 1$, the undamped trajectory may be used inside the integral, since the prefactor in Eq. (18) is already first order in κ_m ; that is, $|\Delta\beta(\tau)|^2 = |\alpha(\tau)|^2 (\Delta S_{\mathbf{s}\mathbf{s}'})^2 [1 + O(\kappa_m \tau)]$. Inserting Eq. (14) into Eq. (18) and integrating (Appendix E) gives $\Lambda(t) \propto t - \sin(\Omega t)/\Omega$. At $t = t_k = 2\pi k/\Omega$ the sine term vanishes and

$$\Lambda(t_k) = \frac{2\pi k \kappa_m (2\bar{n}_{\text{th}} + 1) (\Delta S_{\mathbf{s}\mathbf{s}'})^2}{\Omega^3} + O\left[\left(\frac{\kappa_m}{\Omega}\right)^2\right]. \quad (19)$$

The omitted terms are smaller by a factor of order $\kappa_m t_k$. For longer times the exact form Eq. (E13) applies; its growth rate for $\kappa_m t \gg 1$ is half that of Eq. (E15), because the damped trajectories converge to branch-dependent equilibria separated by $|\Delta S_{\mathbf{s}\mathbf{s}'}|/(\Omega^2 + \kappa_m^2/4)^{1/2}$. Damping changes the entangling phase much less than it

changes the visibility. With damping, the commutator $[H_I(t_1), H_I(t_2)]$ of Appendix D acquires the factor $e^{-\kappa_m(t_1-t_2)/2}$, and the long-time rate at which the phase accumulates changes from $1/\Omega$ to $\Omega/(\Omega^2 + \kappa_m^2/4)$. At closure, θ_k is therefore shifted only by a relative amount of order $(\kappa_m/\Omega)^2$, whereas the visibility is reduced at first order, by $e^{-\Lambda(t_k)}$.

The contrast between Eqs. (15) and (19) is the central result of this section. Initial thermal mixing produces reversible qubit-oscillator dephasing that vanishes identically at unitary closure. Mechanical damping instead exports branch information to the bath, leaving a residual factor $e^{-\Lambda(t_k)}$ that no choice of interrogation time can remove. Since $\Lambda(t_k) \propto \kappa_m \bar{n}_{\text{th}}$ at large occupation, the figure of merit for the mediator is the heating rate $\dot{n} \simeq \kappa_m \bar{n}_{\text{th}}$ rather than the bath occupation alone.

D. Independent qubit dephasing

For a computational-basis matrix element $\rho_{ss'}$, the action of the i th dephasing channel follows from $Z_i|s\rangle = s_i|s\rangle$ (Appendix E). If $s_i = s'_i$ the matrix element is unaffected; if $s_i = -s'_i$ then $s_i s'_i = -1$ and the element decays at rate $\gamma_{\phi,i}$. A coherence differing in the set $\mathcal{F}_{ss'}$ of flipped qubits therefore acquires the factor

$$C_{ss'}^{(\phi)}(t) = \exp \left[-t \sum_{i \in \mathcal{F}_{ss'}} \gamma_{\phi,i} \right], \quad (20)$$

which for equal rates and a coherence differing in both qubits reduces to $e^{-2\gamma_{\phi}t}$. This reproduces from first principles the envelope imposed phenomenologically in Fig. 2, and unlike Eq. (13) it is not periodic: qubit dephasing sets the secular decay of the signal and hence, together with Eq. (19), the largest usable loop index k . For the observable $Y_1 Z_2$ used for readout in Sec. VIB, with Y_i the Pauli- y operator of qubit i , which connects $|s\rangle$ to the state with s_1 reversed, $\mathcal{F}_{ss'} = \{1\}$ and the factor is $e^{-\gamma_{\phi,1}t}$, whereas the coherences in which both qubits are reversed carry $e^{-(\gamma_{\phi,1} + \gamma_{\phi,2})t}$, the envelope of Fig. 2(b).

E. Numerical implementation and convergence

Equation (11) is solved in the Hilbert space $\mathcal{H} = \mathcal{H}_{q_1} \otimes \mathcal{H}_{q_2} \otimes \mathcal{H}_m$ of qubit 1, qubit 2, and the mechanical mode, with the mechanical Hilbert space truncated to the number states $|0\rangle, \dots, |N_b - 1\rangle$. Under column-stacking vectorization, $\text{vec}(A\rho B) = (B^T \otimes A) \text{vec}(\rho)$ for matrices A and B , the master equation becomes $\frac{d}{dt}|\rho\rangle\rangle = \mathcal{L}|\rho\rangle\rangle$, where $|\rho\rangle\rangle = \text{vec}(\rho)$ and $\mathcal{L}$ is the Liouvillian matrix that represents the right-hand side of Eq. (11), with formal solution $|\rho(t)\rangle\rangle = e^{t\mathcal{L}}|\rho(0)\rangle\rangle$; the state is propagated with a sparse exponential-action algorithm. For a thermal state

the probability excluded by the cutoff is

$$\epsilon_{\text{tail}} = \sum_{n=N_b}^{\infty} p_n = \left(\frac{\bar{n}_0}{1 + \bar{n}_0} \right)^{N_b}, \quad (21)$$

where p_n is the thermal occupation probability of the number state $|n\rangle$, Eq. (E1), which for $\bar{n}_0 = 2$ and $N_b = 18$ gives $\epsilon_{\text{tail}} = (2/3)^{18} \simeq 6.77 \times 10^{-4}$; convergence is verified by repeating the calculation with $N_b = 25$. The illustrative parameters are those of Fig. 2, $\omega_0/2\pi = 200$ Hz and $\lambda_1/2\pi = \lambda_2/2\pi = 25$ Hz, together with $\epsilon = 0.02$, $\bar{n}_0 = \bar{n}_{\text{th}} \in \{0, 0.5, 2\}$, $\kappa_m/2\pi = 0.5$ Hz and $\gamma_{\phi,1} = \gamma_{\phi,2} = 0.5$ s $^{-1}$. These values are chosen to display the stroboscopic mechanism within an accessible simulation interval; they are an idealized, platform-agnostic parameter set, and no assignment to a specific experimental platform is implied. In every run the bath occupation is set equal to the initial occupation, $\bar{n}_{\text{th}} = \bar{n}_0$, so the simulations probe the combined effect of the two. Their distinct roles, reversible for $\bar{n}_0$ and irreversible for $\kappa_m \bar{n}_{\text{th}}$ are established analytically in Secs. VB and VC.

The reduced states are $\rho_q = \text{Tr}_m \rho$ and $\rho_m = \text{Tr}_q \rho$. We monitor the qubit-oscillator mutual information $I(q:m) = \mathcal{S}[\rho_q] + \mathcal{S}[\rho_m] - \mathcal{S}[\rho]$ with $\mathcal{S}[\sigma] = -\text{Tr}(\sigma \ln \sigma)$ the von Neumann entropy (distinct from the force operator S), the reduced purity $\mathcal{P}_q = \text{Tr}[\rho_q^2]$, and the mean mechanical occupation $\bar{n}_m(t) = \text{Tr}[b^\dagger b \rho_m(t)]$. The concurrence is evaluated from $\tilde{\rho}_q = (Y_1 Y_2) \rho_q^* (Y_1 Y_2)$, with complex conjugation taken in the computational basis: if $r_1 \geq r_2 \geq r_3 \geq r_4$ are the square roots of the eigenvalues of $\rho_q \tilde{\rho}_q$, then $C[\rho_q] = \max(0, r_1 - r_2 - r_3 - r_4)$ [34], which at closure and in the unitary limit reduces to Eq. (10).

F. Open-system entanglement dynamics

Fig. 3 presents the full solution of Eq. (11). Panel (a) shows that the concurrence is strongly suppressed between closure times, because the qubits become correlated with thermally occupied, spin-dependent mechanical trajectories in accordance with Eq. (13). Near each nominal closure the concurrence revives and approaches the effective ZZ prediction of Eq. (10). The revival is not mathematically exact at finite κ_m , owing both to the residual displacement of Eq. (17) and to the bath-induced factor $e^{-\Lambda(t_k)}$ of Eq. (19). Quantitatively, at the fourth closure ($t_4 \simeq 20.4$ ms) the simulated concurrence reaches $C = 0.969$ for $\bar{n}_0 = \bar{n}_{\text{th}} = 0$ and $C = 0.934$ for $\bar{n}_0 = \bar{n}_{\text{th}} = 2$, against 0.978 for the effective ZZ -plus-dephasing model, i.e. relative revival deficits of 0.9% and 4.5%; for the maximal coherence, Eq. (19) gives $\Lambda(t_1) \simeq 0.021$ at $\bar{n}_{\text{th}} = 2$, a visibility loss of about 2% per loop, consistent with these deficits. Closure-time values are read at the simulation grid point nearest to t_k .

Panel (b) shows the qubit-oscillator mutual information. In the unitary limit, it vanishes exactly at every closure, as required by Eq. (9). With mechanical damping its minima remain close to zero in the weak-damping

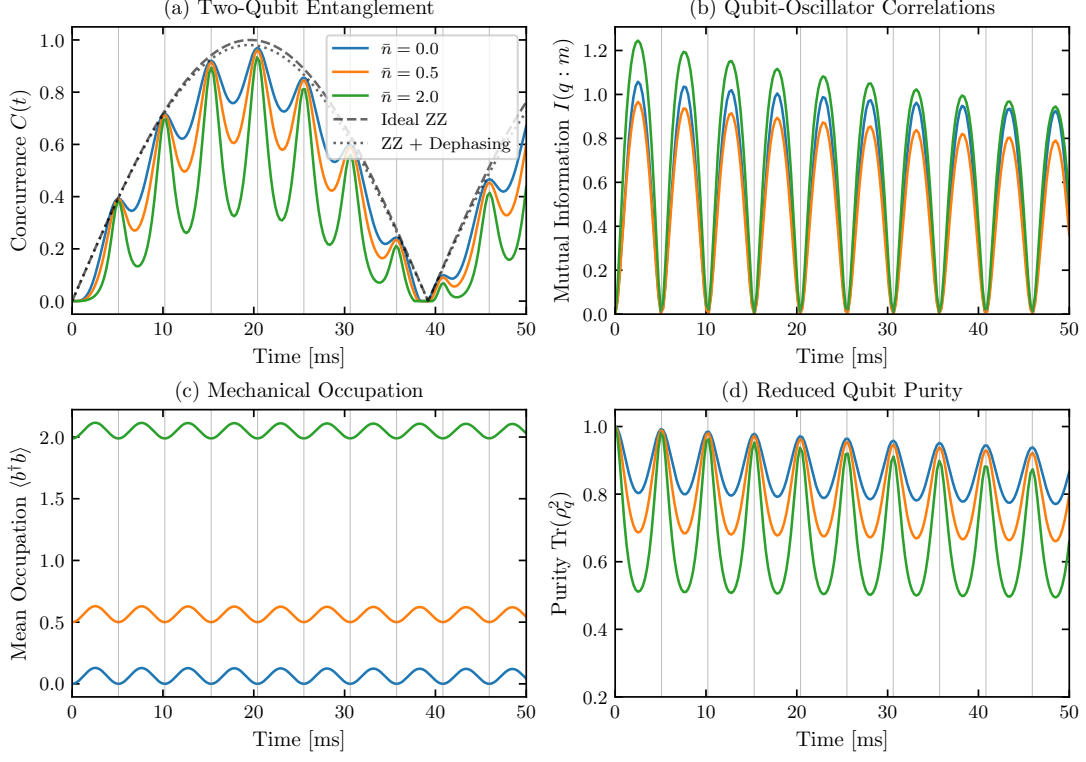


FIG. 3. Open-system dynamics from the full master equation (11), retaining the mechanical mode explicitly. (a) Two-qubit concurrence: strong suppression between closure times due to the thermal factor (13), with revivals at $t_k = 2\pi k/\Omega$ that are exact in the unitary limit and reduced by $e^{-\Lambda(t_k)}$ at finite κ_m and by the dephasing factor of Eq. (20). (b) Qubit-oscillator mutual information $I(q:m)$, which vanishes at every closure for $\kappa_m = 0$. (c) Mean mechanical occupation $\bar{n}_m(t)$, oscillating about the thermal background under the spin-dependent force. (d) Reduced two-qubit purity $\mathcal{P}_q(t)$, showing partial revivals superposed on a secular envelope set by mechanical leakage and qubit dephasing. Parameters as in Sec. VE: $\omega_0/2\pi = 200$ Hz, $\lambda_{1,2}/2\pi = 25$ Hz, $\epsilon = 0.02$, $\bar{n}_0 = \bar{n}_{\text{th}} = 2$, $\kappa_m/2\pi = 0.5$ Hz, $\gamma_{\phi,1} = \gamma_{\phi,2} = 0.5$ s $^{-1}$, and $N_b = 18$ Fock states.

regime, although exact factorization is generally lost at finite κ_m ; the correlations grow between closures as the conditional trajectories separate. Panel (c) displays the mean mechanical occupation, which oscillates about the thermal background because the longitudinal coupling of Eq. (4) produces spin-dependent displacements; the small driven occupation relative to the chosen cutoff, together with the $N_b = 25$ convergence test, confirms that the Fock-space truncation is adequate. Panel (d) shows the reduced two-qubit purity, whose partial revival near the closure times reflects the return of information temporarily stored in the mechanical mode, and whose secular envelope is governed by irreversible mechanical leakage and direct qubit dephasing.

The simulations therefore confirm the central analytic distinction of this section: initial thermal occupation of the mediator is reversible and is undone exactly at the stroboscopic closure times, whereas mechanical damping and qubit dephasing are not. The curvature estimate of Sec. VI is consequently independent of the initial mechanical occupation $\bar{n}_0$ at closure, and is limited instead by the heating rate $\kappa_m \bar{n}_{\text{th}}$ of the bath and by γ_ϕ , both of which enter only through the accumulated interrogation time t_k . These factors multiply the coherences that

carry the parity readout of Sec. VIB; their consequences for the curvature estimate are stated there.

VI. GRAVITATIONAL CURVATURE ESTIMATION

Throughout this section the system is closed and the state is evaluated at exact closure, so that the oscillator is disentangled and the curvature enters only through the phase $\theta_k(\Gamma_{\text{eq}})$; the extension of the estimation analysis to the open-system state of Sec. V is left for future work.

A. Quantum Fisher information

The curvature Γ_{eq} is encoded via the phase $\theta_k(\Gamma_{\text{eq}})$. For the input state $|++\rangle$, the quantum Fisher information (QFI) is (Appendix F)

$$F_Q = 4 \left(\partial_{\Gamma_{\text{eq}}} \theta_k \right)^2, \quad (22)$$

and the quantum Cramér-Rao bound on ν independent shots is $\text{Var}(\hat{\Gamma}) \geq 1/(\nu F_Q)$. For stabilized couplings, F_Q

grows as $(\omega_0^2 - 2\Gamma)^{-4}$ as the mode softens; for microscopic displacement coupling as $(\omega_0^2 - 2\Gamma)^{-5}$ as the modes softens. Eq. (22) is the QFI of a single phase generated by $Z_1 Z_2$, whose invariance in $|++\rangle$ is unity (Appendix F; it is independent of θ_k , so the amount of entanglement present at a given closure time does not determine sensitivity. The role of the two-qubit encoding is selectivity, only the nonlocal $Z_1 Z_2$ phase responds to Γ_{eq} , whereas the local phases respond to g_R and are removed by the echo sequence of Sec. VI C.

The quantum Fisher information quantifies the maximum information about the curvature parameter Γ that is encoded in the two-qubit state and establishes the ultimate precision permitted by quantum mechanics. However, attaining this limit requires an experimentally accessible measurement whose statistical sensitivity matches the QFI. It is therefore necessary to identify an observable that converts the curvature-dependent phase $\theta_k(\Gamma)$ into a measurable signal while preserving the full metrological advantage of the state. In the following subsection, we show that a two-qubit parity measurement provides precisely such a readout, yielding a classical Fisher information equal to the quantum Fisher information and thereby saturating the quantum Cramér-Rao bound.

B. Parity measurement

The observable $Y_1 Z_2$, with Y_1 the Pauli-y operator of qubit 1, satisfies (Appendix F) $\langle Y_1 Z_2 \rangle_\Gamma = -\sin(2\theta_k)$, where $\langle \cdot \rangle_\Gamma$ is the expectation value in the output state $|\psi_\Gamma\rangle = e^{i\theta_k Z_1 Z_2} |++\rangle$, and its classical Fisher information equals F_Q exactly. The two outcomes have probabilities $p_\pm = [1 \mp \sin(2\theta_k)]/2$, and the measurement saturates the quantum Cramér-Rao bound at every operating point for the ideal closed-system state and an ideal projective readout. Experimentally, a local $\pi/2$ pulse about x on qubit 1 before standard two-qubit parity readout implements $Y_1 Z_2$. The analytical results hold for the closed system. Qubit dephasing is included phenomenologically in Fig. (2) through the envelope $e^{-2\gamma_\phi t}$, which reduces visibility of the parity fringe and therefore the information obtained per shot. Since θ_k grows with the number of loops k while the envelope decays with t_k , the number of loops over which the entangling phase can usefully be accumulated is limited by γ_ϕ , the pure dephasing rate. Mechanical dissipation is likewise not included; closure at t_k is independent of the initial mechanical state, so the thermal occupation of the shared mode does not affect the stroboscopic gate, but damping of the oscillator during a loop prevents trajectories from closing exactly and leaves residual qubit-oscillator correlations at t_k .

C. Differential Source Protocol

The parity measurement described above provides an optimal readout of the curvature-dependent phase θ_k , but practical sensing also requires the suppression of unwanted backgrounds that do not originate from the tidal field. In particular, the uniform gravitational acceleration produced by the source mass generates local single-qubit phase shifts through the force channel. These phases do not carry information about the curvature and can mask the desired entangling signal if left uncompensated.

To isolate the curvature contribution, we employ simultaneous π pulses on both qubits, applied at stroboscopic closure times, which reverse the sign of the local Z_i evolution while preserving the commuting two-qubit interaction $Z_1 Z_2$ and the closed phase-space trajectories. Since the effective entangling Hamiltonian is proportional to $Z_1 Z_2$, echo pulses cancel single-body force-induced phases without removing the accumulated entangling phase θ_k . Both requirements are necessary; a π pulse on one qubit alone maps $Z_1 Z_2 \rightarrow -Z_1 Z_2$ and would cancel the entangling phase together with the local one, and a pulse applied away from a closure time flips the sign of S in Eq. (8), so that the trajectory no longer closes. Applied at t_k , the pulse pair leaves S^2 , and hence θ_k , unchanged while reversing all local terms, so that static local phases cancel over a symmetric sequence. The protocol, therefore, selectively suppresses the force channel while maintaining sensitivity to curvature-induced modifications of the oscillator-mediated interaction.

Additional discrimination is obtained through differential source mass modulation. By cycling the source mass between two positions, R_A and R_B , the experiment measures a curvature difference

$$\Delta\Gamma = GM (R_A^{-3} - R_B^{-3}), \quad (23)$$

while the dominant force background varies as R^{-2} . Because the curvature and force channels possess different distance scalings, modulating the source position provides a form of spectral and geometric separation between the desired tidal signal and residual acceleration-induced backgrounds.

The resulting differential entangling phase is

$$\Delta\phi_k = 2 (\partial_\Gamma \theta_k) \Delta\Gamma, \quad (24)$$

which directly converts a change in gravitational curvature into a measurable shift of the parity fringe, whose phase is $2\theta_k$; $\Delta\phi_k$ is the stroboscopic counterpart of the fixed-time phase difference $\Delta\phi$ of Fig. 2(c). The characteristic R^{-3} dependence of $\Delta\phi_k$ serves as a distinctive signature of a point-source tidal field, providing an experimental consistency check that distinguishes the curvature signal from force-channel effects and other slowly varying backgrounds. In this way, the combination of

parity readout, spin echo, and differential source modulation forms a complete sensing protocol that approaches the quantum limit identified by the quantum Fisher information analysis.

Three points complete the protocol. First, Ω differs between the two source positions, so the closure time $t_k = 2\pi k/\Omega$ must be set separately at R_A and R_B . Second, only the differential measurement is practicable. An absolute estimate of Γ_{eq} from θ_k would require $\lambda_1\lambda_2$ and ω_0 to be known to better than the fractional curvature shift itself, 2ϵ or 3ϵ (Sec. IV); source modulation instead turns slow drifts of λ_i and ω_0 into a common-mode background. Third, the results hold within the model of Sec. II: a harmonic, one-dimensional oscillator, a static pointlike or spherical source, and purely longitudinal coupling. Damping and dephasing are treated in Sec. V, where they are shown to limit the largest usable loop index k , while the thermal occupation of the mediator cancels at closure. The parameters of Figs. 2 and 3 are illustrative and are not a detectability forecast. This work therefore establishes the exact transduction chain of Eq. (7) and its metrological characterization under Markovian noise, not feasibility on a specific platform.

VII. CONCLUSION

We have presented a quantum-metrological framework in which gravitational curvature is converted into a measurable two-qubit entangling phase through its influence on a shared mechanical oscillator. The central mechanism is the curvature-induced renormalization of the mechanical frequency, Eq. (3), which modifies the susceptibility $\chi_m = 1/m\Omega^2$ of the oscillator and thereby controls the strength of the mediated Ising interaction between two longitudinally coupled qubits. This establishes the explicit transduction chain of Eq. (7), running from the gravitational tidal field to a nonlocal quantum correlation.

The qubit-oscillator dynamics were treated exactly. An exact polaron transformation eliminates the qubit-oscillator interaction without requiring rotating-wave, dispersive, or weak-coupling approximations, yielding the effective ZZ interaction of Eq. (5) whose strength Eq. (6) depends directly on the curvature-renormalized mechanical frequency. In the laboratory frame, an exactly terminated Magnus expansion gives the propagator Eq. (8) and reveals spin-dependent closed trajectories in mechanical phase space. At the stroboscopic closure times $t_k = 2\pi k/\Omega$ the oscillator disentangles completely from the qubits, leaving the pure entangling gate Eq. (9) with $\theta_k = Jt_k$.

Starting from an initially separable state, the accumulated phase generates entanglement quantified by the concurrence of Eq. (10), $C(t_k) = |\sin(2\theta_k)| = |\sin(2Jt_k)|$, demonstrating that the gravitational curvature is directly encoded in an experimentally observ-

able two-qubit correlation. Because both the interaction strength J and the closure time t_k inherit the curvature dependence of the mechanical frequency, the resulting entangling phase provides a natural parameter through which the tidal field can be estimated, and the closure condition itself contributes to the sensitivity: $\theta_k \propto \Omega^{-2}$ for stabilized couplings and Ω^{-3} for microscopic displacement coupling, against Ω^{-1} and Ω^{-2} for a phase accumulated over a fixed time.

Treating the curvature as an unknown parameter, we derived the quantum Fisher information Eq. (22) associated with the encoded phase and established the corresponding quantum Cramér-Rao bound. We further identified a two-qubit parity measurement whose classical Fisher information equals the quantum Fisher information, thereby providing an optimal and experimentally accessible readout strategy. The introduction of a spin-echo-assisted differential source protocol enables suppression of force-channel backgrounds while preserving sensitivity to the curvature-dependent entangling phase, allowing the distinctive R^{-3} tidal scaling of the signal to be isolated from the R^{-2} acceleration background. These results hold for the closed system at closure; the Fisher information analysis of the open-system state is left for future work.

The open-system analysis of Sec. V sharpens what is and is not required of the mediator. Solving the full master equation Eq. (11) with the mechanical mode retained explicitly, we showed that an initially thermal oscillator attenuates the qubit coherences by the factor Eq. (13), which is periodic in the conditional displacement and returns to unity exactly at every closure time, Eq. (15), for arbitrary initial occupation $\bar{n}_0$. Thermal population of the mediator is therefore reversible, and ground-state cooling is not a prerequisite for the protocol; the entangling gate, the concurrence, and the quantum Fisher information are all independent of the initial mechanical temperature at closure. Mechanical damping is qualitatively different: it exports branch information irreversibly to the reservoir, deforming the conditional trajectories so that the nominal closure leaves the residual displacement Eq. (17) and imprinting the visibility factor $e^{-\Lambda(t_k)}$ of Eq. (19), which no choice of interrogation time can remove. Together with the qubit dephasing envelope Eq. (20), this identifies the heating rate $\dot{n} \simeq \kappa_m \bar{n}_{\text{th}}$ and the dephasing rate γ_ϕ , rather than the bath occupation alone, as the figures of merit that limit the usable loop index k and hence the attainable precision.

The scheme proposed here is conceptually distinct from gravity-mediated entanglement proposals. The mechanical oscillator serves as the explicit quantum mediator, while gravity acts solely as a classical perturbation that modifies the properties of that mediator through the tidal field. Consequently, the observation of entanglement in this setting constitutes evidence of gravitational control of a quantum interaction rather than entanglement generated by a quantized gravitational field.

Several extensions follow naturally. The trace-free

structure of the Newtonian tidal tensor (Appendix B) stiffens the transverse modes by $+\Gamma_{\text{eq}}$ while softening the radial mode by $-2\Gamma_{\text{eq}}$, so a two-oscillator architecture reading out both mode families differentially would exploit a tensor signature unavailable to scalar force sensors. Because closure must be timed at the curvature-shifted frequency $\Omega(\Gamma_{\text{eq}})$ rather than at ω_0 , adaptive protocols that track the closure condition itself constitute a second estimation channel worth quantifying. Finally, the analysis assumed a static source and the harmonic truncation of Eq. (3); extending it to a modulated source mass, and to the anharmonic corrections that become relevant when the mechanical amplitude is no longer small compared with R , would connect the present framework more closely to the operating regimes of levitated and soft-clamped resonators. More broadly, the framework demonstrates how weak gravitational curvature can be converted into a nonlocal quantum resource and exploited for precision sensing using hybrid quantum systems.

DATA AVAILABILITY

No data were created or analyzed in this study. All results are analytic and are reproducible from the expressions in the paper together with the numerical verification.

Appendix A: Gravitational Renormalization of the Mechanical Mode

1. Newtonian expansion

For $|x| \ll R$ the exact potential $V_G(x) = -GMm/(R+x)$ expands as

$$V_G(x) = -\frac{GMm}{R} \sum_{n=0}^{\infty} (-1)^n \left(\frac{x}{R}\right)^n, \quad (\text{A1})$$

convergent for $|x| < R$. Retaining through x^2 and adding the bare harmonic potential gives $V_{\text{tot}}^{(2)}(x) = V_0 + mg_R x + \frac{m}{2}(\omega_0^2 - 2\Gamma_R)x^2$, with V_0 constant, $g_R = GM/R^2$, and $\Gamma_R = GM/R^3$.

2. Equilibrium position

Setting $dV_{\text{tot}}/dx = 0$ for the full potential $V_{\text{tot}}(x) = \frac{m}{2}\omega_0^2 x^2 + V_G(x)$ gives $u + \zeta/(1+u)^2 = 0$, with $u = x_{\text{eq}}/R$ and $\zeta = \Gamma_R/\omega_0^2$. Iterating $u = -\zeta(1+u)^{-2}$ for $|\zeta| \ll 1$,

$$x_{\text{eq}} = -\frac{g_R}{\omega_0^2} \left[1 + 2\frac{\Gamma_R}{\omega_0^2} + 7\left(\frac{\Gamma_R}{\omega_0^2}\right)^2 + \mathcal{O}(\zeta^3) \right], \quad (\text{A2})$$

so $x_{\text{eq}} \simeq -g_R/\omega_0^2$ at leading order.

Appendix B: Tidal Tensor in Three Dimensions

The Newtonian tidal tensor $\Gamma_{ij} = -\partial_i \partial_j \Phi_G$, with $\Phi_G = -GM/r$ the potential at distance r from the source, has components $\Gamma_{rr} = 2GM/R^3$ and $\Gamma_{\perp\perp} = -GM/R^3$ at distance R from a point source, and is trace-free, $\Gamma_{rr} + 2\Gamma_{\perp\perp} = 0$. The linearized equation of motion along direction i is $\ddot{x}_i = -\omega_0^2 x_i + \Gamma_{ii} x_i$, so a positive (negative) diagonal component softens (stiffens) the mode. Radially, $\Gamma_{\text{eq}} = \Gamma_{rr}/2 = GM/R_{\text{eq}}^3$ with $R_{\text{eq}} = R + x_{\text{eq}}$, giving $\Omega_r^2 = \omega_0^2 - 2\Gamma_{\text{eq}}$; transversely, $\Omega_{\perp}^2 = \omega_0^2 + \Gamma_{\text{eq}}$. The spectrum therefore splits into one soft and two stiff modes, and the main text treats the soft radial mode. The transverse shift is $+\Gamma_{\text{eq}}$ against $-2\Gamma_{\text{eq}}$ for the radial mode, so a differential radial-transverse readout carries $3\Gamma_{\text{eq}}$, a factor $3/2$ over the radial mode alone [2].

Appendix C: Exact Polaron Transformation

Define $A = S/\Omega$ and $K = (b^\dagger - b)A$, so that $U_P = e^K$ is the displacement operator $D[A]$ of Eq. (8) with operator-valued amplitude A , which commutes with b , $b^\dagger$, and itself. The only nonvanishing commutant is $[K, b] = -A$, since $[K, [K, b]] = -[K, A] = 0$, so $U_P b U_P^\dagger = b - A$ and $U_P b^\dagger U_P^\dagger = b^\dagger - A$. Applying U_P to Eq. (4) in units of $\hbar$, with the qubit term $\sum_i \omega_{q,i} Z_i/2$ invariant, and using $\Omega A = S$:

$$\begin{aligned} U_P(\Omega b^\dagger b) U_P^\dagger &= \Omega b^\dagger b - \Omega A(b + b^\dagger) + \Omega A^2, \\ U_P[S(b + b^\dagger)] U_P^\dagger &= S(b + b^\dagger) - 2SA. \end{aligned}$$

The terms linear in $b + b^\dagger$ cancel exactly. With $SA = S^2/\Omega$ and $S^2 = (\lambda_1^2 + \lambda_2^2)\mathbf{1} + 2\lambda_1\lambda_2 Z_1 Z_2$, the residual is $-S^2/\Omega = -(\lambda_1^2 + \lambda_2^2)/\Omega \mathbf{1} - J Z_1 Z_2$, whose second term is Eq. (6).

Appendix D: Magnus Expansion and Exact Propagator

The interaction-picture Hamiltonian is $H_I(t) = \hbar S(b e^{-i\Omega t} + b^\dagger e^{i\Omega t})$, and $U_I(t) = \exp(\mathcal{M}_1 + \mathcal{M}_2 + \dots)$ with $\mathcal{M}_1 = -(i/\hbar) \int_0^t H_I dt'$ and $\mathcal{M}_2 = -(1/2\hbar^2) \int_0^t dt_1 \int_0^{t_1} dt_2 [H_I(t_1), H_I(t_2)]$ [28, 29]. The first term is $\mathcal{M}_1(t) = S[\alpha(t)b^\dagger - \alpha^*(t)b]$ with $\alpha(t) = (1 - e^{i\Omega t})/\Omega$. The commutator $[H_I(t_1), H_I(t_2)] = -2i\hbar^2 S^2 \sin[\Omega(t_1 - t_2)]$ is a c-number times S^2 , which commutes with $H_I(t)$ at all times, so all terms of order $n \geq 3$ vanish and $\mathcal{M}_2(t) = i\Phi(t)S^2$ with $\Phi(t) = (\Omega t - \sin \Omega t)/\Omega^2$. Since $[\mathcal{M}_1, \mathcal{M}_2] = 0$, the propagator factorizes as Eq. (8). Because $U_I(t_k)$ depends on S^2 alone, the echo pulses of Sec. VIC are applied only at closure times.

At closure, $\alpha(t_k) = 0$ and $\Phi(t_k) = 2\pi k/\Omega^2$, so $U_I(t_k) = e^{i\Phi(t_k)S^2}$. Expanding S^2 gives the

global phase $e^{i\Phi(t_k)(\lambda_1^2+\lambda_2^2)}$ and the entangling operation $e^{2i\lambda_1\lambda_2\Phi(t_k)Z_1Z_2} = e^{i\theta_k Z_1Z_2}$ with $\theta_k = 4\pi k\lambda_1\lambda_2/\Omega^2 = Jt_k$.

Appendix E: Derivations for the Open-System Dynamics

1. Thermal initial state and reduced qubit dynamics

The thermal state of Sec. V B is

$$\rho_m^{\text{th}}(\bar{n}_0) = \sum_{n=0}^{\infty} p_n |n\rangle\langle n|, \quad p_n = \frac{1}{1+\bar{n}_0} \left(\frac{\bar{n}_0}{1+\bar{n}_0} \right)^n. \quad (\text{E1})$$

The propagator of Eq. (8) acts on the simultaneous eigenstates $|s\rangle = |s_1, s_2\rangle$ of Z_1 and Z_2 , which diagonalize $S = \lambda_1 Z_1 + \lambda_2 Z_2$ with eigenvalues $S_s = s_1\lambda_1 + s_2\lambda_2$, as

$$U_I(t)|s\rangle = e^{i\Phi(t)S_s^2} D[\alpha(t)S_s] |s\rangle. \quad (\text{E2})$$

With $\rho_q(0) = \sum_{s,s'} \rho_{ss'}^{(q)}(0) |s\rangle\langle s'|$ and $\rho_I(t) = U_I(t)\rho(0)U_I^\dagger(t)$,

$$\begin{aligned} \rho_I(t) &= \sum_{s,s'} \rho_{ss'}^{(q)}(0) e^{i\Phi(t)(S_s^2 - S_{s'}^2)} |s\rangle\langle s'| \\ &\otimes D(\alpha S_s) \rho_m^{\text{th}} D^\dagger(\alpha S_{s'}). \end{aligned} \quad (\text{E3})$$

Tracing out the oscillator gives

$$\rho_{ss'}^{(q)}(t) = \rho_{ss'}^{(q)}(0) e^{i\Phi(t)(S_s^2 - S_{s'}^2)} \chi_{ss'}(t), \quad (\text{E4})$$

with, by cyclicity of the trace,

$$\chi_{ss'}(t) = \text{Tr}_m[D^\dagger(\alpha S_{s'}) D(\alpha S_s) \rho_m^{\text{th}}]. \quad (\text{E5})$$

The Weyl composition relation $D(\xi_1)D(\xi_2) = \exp[(\xi_1\xi_2^* - \xi_1^*\xi_2)/2] D(\xi_1 + \xi_2)$ with $D^\dagger(\xi) = D(-\xi)$ gives

$$\begin{aligned} D^\dagger(\alpha S_{s'}) D(\alpha S_s) &= \exp\left[\frac{-\alpha S_{s'} \alpha^* S_s + \alpha^* S_{s'} \alpha S_s}{2}\right] \\ &\times D[\alpha \Delta S_{ss'}], \end{aligned} \quad (\text{E6})$$

where $\Delta S_{ss'} \equiv S_s - S_{s'}$. Since S_s and $S_{s'}$ are real the exponent vanishes identically, so $\chi_{ss'}(t) = \text{Tr}_m[D(\xi)\rho_m^{\text{th}}]$ with $\xi = \alpha(t)\Delta S_{ss'}$. Substituting Eq. (E1) and using $\langle n|D(\xi)|n\rangle = e^{-|\xi|^2/2} L_n(|\xi|^2)$, with L_n the Laguerre polynomial,

$$\text{Tr}_m[D(\xi)\rho_m^{\text{th}}] = \frac{e^{-|\xi|^2/2}}{1+\bar{n}_0} \sum_{n=0}^{\infty} \left(\frac{\bar{n}_0}{1+\bar{n}_0} \right)^n L_n(|\xi|^2). \quad (\text{E7})$$

The generating function $\sum_{n \geq 0} r^n L_n(x) = (1-r)^{-1} \exp[-rx/(1-r)]$, valid for $|r| < 1$, evaluated at $r = \bar{n}_0/(1+\bar{n}_0)$, for which $1-r = (1+\bar{n}_0)^{-1}$ and $r/(1-r) = \bar{n}_0$, gives $\text{Tr}_m[D(\xi)\rho_m^{\text{th}}] = e^{-|\xi|^2/2} e^{-\bar{n}_0|\xi|^2} = \exp[-\frac{1}{2}(2\bar{n}_0+1)|\xi|^2]$, which with $\xi = \alpha(t)\Delta S_{ss'}$ is Eq. (13).

2. Conditional oscillator amplitude under damping

The conditional first moment $\beta_s(t)$ of Sec. V C follows from Eq. (11). Using $[b, b^\dagger b] = b$ and $[b, b + b^\dagger] = 1$, the Hamiltonian commutator contributes $-i\Omega\beta_s - iS_s$ and the two mechanical dissipators $-(\kappa_m/2)\beta_s$, so

$$\dot{\beta}_s = -\left(i\Omega + \frac{\kappa_m}{2}\right)\beta_s - iS_s. \quad (\text{E8})$$

Subtracting the two instances of Eq. (E8) with $\Delta\beta = \beta_s - \beta_{s'}$,

$$\frac{d}{dt}\Delta\beta = -\left(i\Omega + \frac{\kappa_m}{2}\right)\Delta\beta - i\Delta S_{ss'}. \quad (\text{E9})$$

For any initial mechanical state common to both branches, $\Delta\beta(0) = 0$, and the integrating factor $\exp[(i\Omega + \kappa_m/2)t]$ gives

$$\Delta\beta(t) = -i\Delta S_{ss'} \frac{1 - e^{-(i\Omega + \kappa_m/2)t}}{i\Omega + \kappa_m/2}, \quad (\text{E10})$$

whose squared magnitude is Eq. (16).

3. Information leakage to the bath

Over an interval dt , mechanical damping is a thermal attenuation channel of transmissivity $\mathcal{T}_{dt} = e^{-\kappa_m dt} = 1 - \kappa_m dt + O(dt^2)$. In a Stinespring realization the mode is mixed with an independent thermal bath mode on a beam splitter, so the amplitudes β_s and $\beta_{s'}$ generate bath amplitudes separated by

$$\delta\zeta = \sqrt{1 - \mathcal{T}_{dt}} \Delta\beta(t) = \sqrt{\kappa_m dt} \Delta\beta(t) + O(dt^{3/2}). \quad (\text{E11})$$

The bath overlap $\mathcal{V}_{dt}$ is the thermal characteristic function evaluated above with $\bar{n}_0 \rightarrow \bar{n}_{\text{th}}$ and $\xi \rightarrow \delta\zeta$:

$$\mathcal{V}_{dt} = \exp\left[-\frac{\kappa_m}{2}(2\bar{n}_{\text{th}} + 1)|\Delta\beta(t)|^2 dt + O(dt^2)\right]. \quad (\text{E12})$$

Independent bath increments multiply, so $\prod_j \mathcal{V}_{dt_j} = e^{-\Lambda(t)}$ with $\Lambda(t)$ given by Eq. (18).

On the exact damped path, writing $a = \kappa_m/2$ and $d = \Omega^2 + a^2$, substitution of Eq. (16) requires $\int_0^t d\tau = t$, $\int_0^t e^{-2a\tau} d\tau = (1 - e^{-\kappa_m t})/\kappa_m$, and $\int_0^t e^{-a\tau} \cos \Omega\tau d\tau = \{a + e^{-at}[-a \cos \Omega t + \Omega \sin \Omega t]\}/d$, giving

$$\begin{aligned} \Lambda_d(t) &= \frac{\kappa_m}{2} (2\bar{n}_{\text{th}} + 1) \frac{(\Delta S_{ss'})^2}{d} \left\{ t + \frac{1 - e^{-\kappa_m t}}{\kappa_m} \right. \\ &\quad \left. - \frac{2}{d} \left[a + e^{-at}(-a \cos \Omega t + \Omega \sin \Omega t) \right] \right\}. \end{aligned} \quad (\text{E13})$$

In the weak-damping regime of Sec. V C, $|\Delta\beta(\tau)|^2$ is replaced by $|\alpha(\tau)|^2 (\Delta S_{ss'})^2$ in Eq. (18). With Eq. (14),

$$\int_0^t |\alpha(\tau)|^2 d\tau = \frac{2}{\Omega^2} \int_0^t [1 - \cos \Omega\tau] d\tau = \frac{2}{\Omega^2} \left[t - \frac{\sin \Omega t}{\Omega} \right], \quad (\text{E14})$$

so the leading-order damping exponent at arbitrary time is

$$\Lambda(t) = \frac{\kappa_m(2\bar{n}_{\text{th}} + 1)(\Delta S_{ss'})^2}{\Omega^2} \left[t - \frac{\sin \Omega t}{\Omega} \right] + O\left[\left(\frac{\kappa_m}{\Omega}\right)^2\right], \quad (\text{E15})$$

which at $t_k = 2\pi k/\Omega$ reduces to Eq. (19).

4. Qubit dephasing

Since $Z_i^2 = \mathbf{1}$, the qubit dissipator is

$$\frac{\gamma_{\phi,i}}{2} \mathcal{D}[Z_i]\rho = \frac{\gamma_{\phi,i}}{2} (Z_i \rho Z_i - \rho), \quad (\text{E16})$$

which for a computational-basis element $\rho_{ss'}$, with $Z_i|\mathbf{s}\rangle = s_i|\mathbf{s}\rangle$, gives

$$\left. \frac{d}{dt} \rho_{ss'} \right|_{\phi,i} = \frac{\gamma_{\phi,i}}{2} (s_i s'_i - 1) \rho_{ss'}. \quad (\text{E17})$$

Integrating and multiplying the factors of the channels with $s_i \neq s'_i$ gives Eq. (20).

Appendix F: Quantum Fisher Information and Classical Fisher Saturation

For the pure output state $|\psi_\Gamma\rangle = e^{i\theta_k Z_1 Z_2} |++\rangle$ the local generator, defined by $\partial_\Gamma |\psi_\Gamma\rangle = -i\mathcal{G}|\psi_\Gamma\rangle$, is $\mathcal{G} = -(\partial_\Gamma \theta_k) Z_1 Z_2$, and $F_Q = 4 \text{Var}(\mathcal{G})$. Since $\langle ++ | Z_1 Z_2 | ++ \rangle = 0$ and $(Z_1 Z_2)^2 = \mathbf{1}$, the variance is $\text{Var}_{++}(Z_1 Z_2) = 1$, giving $F_Q = 4(\partial_\Gamma \theta_k)^2$.

The observable $Y_1 Z_2$ anticommutes with $Z_1 Z_2$, $\{Z_1 Z_2, Y_1 Z_2\} = 0$, since $\{Z_1, Y_1\} = 0$ and $Z_2^2 = \mathbf{1}$. Using $e^{-i\theta\mathcal{A}} \mathcal{B} e^{i\theta\mathcal{A}} = \mathcal{B} \cos(2\theta) - i\mathcal{A}\mathcal{B} \sin(2\theta)$ for anticommuting Hermitian $\mathcal{A}, \mathcal{B}$ with $\mathcal{A}^2 = \mathcal{B}^2 = \mathbf{1}$, here $\mathcal{A} = Z_1 Z_2$, $\mathcal{B} = Y_1 Z_2$, $\theta = \theta_k$, and $-i(Z_1 Z_2)(Y_1 Z_2) = -iZ_1 Y_1 = -X_1$ with X_1 the Pauli- x operator of qubit 1:

$$\langle Y_1 Z_2 \rangle_\Gamma = \langle ++ | [Y_1 Z_2 \cos(2\theta_k) - X_1 \sin(2\theta_k)] | ++ \rangle = -\sin(2\theta_k).$$

The binary probabilities are $p_\pm = (1 \mp \sin(2\theta_k))/2$, and $\sum_r (\partial_\Gamma p_r)^2 / p_r = 4(\partial_\Gamma \theta_k)^2 = F_Q$, where $r = \pm$ labels the two parity outcomes.

-
- [1] M. Aspelmeyer, T. J. Kippenberg, and F. Marquardt, Cavity optomechanics, *Reviews of Modern Physics* **86**, 1391 (2014).
 - [2] C. L. Degen, F. Reinhard, and P. Cappellaro, Quantum sensing, *Reviews of modern physics* **89**, 035002 (2017).
 - [3] C. Gonzalez-Ballester, M. Aspelmeyer, L. Novotny, R. Quidant, and O. Romero-Isart, Levitodynamics: Levitation and control of microscopic objects in vacuum, *Science* **374**, eabg3027 (2021).
 - [4] T. Westphal, H. Hepach, J. Pfaff, and M. Aspelmeyer, Measurement of gravitational coupling between millimetre-sized masses, *Nature* **591**, 225 (2021).
 - [5] A. D. O'Connell, M. Hofheinz, M. Ansmann, R. C. Bialczak, M. Lenander, E. Lucero, M. Neeley, D. Sank, H. Wang, M. Weides, J. Wenner, J. M. Martinis, and A. N. Cleland, Quantum ground state and single-phonon control of a mechanical resonator, *Nature* **464**, 697 (2010).
 - [6] M. Kasevich and S. Chu, Atomic interferometry using stimulated Raman transitions, *Phys. Rev. Lett.* **67**, 181 (1991).
 - [7] S. Qvarfort, A. Serafini, P. F. Barker, and S. Bose, Gravimetry through non-linear optomechanics, *Nat. Commun.* **9**, 3690 (2018).
 - [8] F. Armata, L. Latmiral, A. D. K. Plato, and M. S. Kim, Quantum limits to gravity estimation with optomechanics, *Phys. Rev. A* **96**, 043824 (2017).
 - [9] V. Giovannetti, S. Lloyd, and L. Maccone, Advances in quantum metrology, *Nat. Photonics* **5**, 222 (2011).
 - [10] S. L. Braunstein and C. M. Caves, Statistical distance and the geometry of quantum states, *Physical Review Letters* **72**, 3439 (1994).
 - [11] M. G. Paris, Quantum estimation for quantum technology, *International Journal of Quantum Information* **7**, 125 (2009).
 - [12] B. Royer, A. L. Grimsmo, N. Didier, and A. Blais, Fast and high-fidelity entangling gate through parametrically modulated longitudinal coupling, *Quantum* **1**, 11 (2017).
 - [13] N. Didier, J. Bourassa, and A. Blais, Fast quantum non-demolition readout by parametric modulation of longitudinal qubit-oscillator interaction, *Physical review letters* **115**, 203601 (2015).
 - [14] S. Bose, A. Mazumdar, G. W. Morley, H. Ulbricht, M. Toroš, M. Paternostro, A. A. Geraci, P. F. Barker, M. Kim, and G. Milburn, Spin entanglement witness for quantum gravity, *Physical review letters* **119**, 240401 (2017).
 - [15] C. Marletto and V. Vedral, Gravitationally induced entanglement between two massive particles is sufficient evidence of quantum effects in gravity, *Physical review letters* **119**, 240402 (2017).
 - [16] A. Matsumura and K. Yamamoto, Gravity-induced entanglement in optomechanical systems, *Phys. Rev. D* **102**, 106021 (2020).
 - [17] T. Krisnanda, G. Y. Tham, M. Paternostro, and T. Paterek, Observable quantum entanglement due to gravity, *npj Quantum Inf.* **6**, 12 (2020).
 - [18] O. Bengyat, A. Di Biagio, M. Aspelmeyer, and M. Christodoulou, Gravity-mediated entanglement between oscillators as quantum superposition of geometries, *Phys. Rev. D* **110**, 056046 (2024).
 - [19] Z. Tang, H. Xue, Z. Han, Z. Kan, Z. Li, and Y. Liu, Optimal form factors for experimental proposals on gravity-induced entanglement, *Phys. Rev. D* **112**, 042004 (2025).
 - [20] M. Toroš, A. Mazumdar, and S. Bose, Loss of coherence and coherence protection from a graviton bath, *Physical*

- Review D **109**, 084050 (2024).
- [21] D. Miki, A. Matsumura, and K. Yamamoto, Quantum signature of gravity in optomechanical systems with conditional measurement, *Physical Review D* **109**, 064090 (2024).
 - [22] A. Ali, S. Al-Kuwari, M. Ghominejad, M. T. Rahim, D. Wang, and S. Haddadi, Quantum characteristics near event horizons, *Phys. Rev. D* **110**, 064001 (2024), arXiv:2401.12028.
 - [23] C. M. Will, The confrontation between general relativity and experiment, *Living Rev. Relativ.* **17**, 4 (2014).
 - [24] C. W. Misner, K. S. Thorne, and J. A. Wheeler, *Gravitation* (W. H. Freeman, San Francisco, 1973).
 - [25] E. L. Hahn, Spin echoes, *Phys. Rev.* **80**, 580 (1950).
 - [26] A. Ali, S. Al-Kuwari, M. Hussain, T. Byrnes, M. Rahim, J. Q. Quach, M. Ghominejad, and S. Haddadi, Ergotropy and capacity optimization in heisenberg spin-chain quantum batteries, *Physical Review A* **110**, 052404 (2024).
 - [27] A. Ali, S. Al-Kuwari, and S. Haddadi, Trade-off relations of quantum resource theory in heisenberg models, arXiv preprint arXiv:2401.01063 (2024).
 - [28] W. Magnus, On the exponential solution of differential equations for a linear operator, *Commun. Pure Appl. Math.* **7**, 649 (1954).
 - [29] S. Blanes, F. Casas, J. A. Oteo, and J. Ros, The Magnus expansion and some of its applications, *Phys. Rep.* **470**, 151 (2009).
 - [30] A. Sørensen and K. Mølmer, Entanglement and quantum computation with ions in thermal motion, *Phys. Rev. A* **62**, 022311 (2000).
 - [31] D. Leibfried, B. DeMarco, V. Meyer, D. Lucas, M. Barrett, J. Britton, W. M. Itano, B. Jelenković, C. Langer, T. Rosenband, and D. J. Wineland, Experimental demonstration of a robust, high-fidelity geometric two ion-qubit phase gate, *Nature* **422**, 412 (2003).
 - [32] M. I. Hussain, E. O. Ilo-Okeke, and T. Byrnes, Geometric phase gate for entangling two bose-einstein condensates, *Physical Review A* **89**, 053607 (2014).
 - [33] M. I. Hussain, E. O. Ilo-Okeke, and T. Byrnes, Geometric phase gate based on the ac stark shift, *Quantum Information Processing* **14**, 943 (2015).
 - [34] W. K. Wootters, Entanglement of formation of an arbitrary state of two qubits, *Physical Review Letters* **80**, 2245 (1998).